

```
1 clc;
2 clear;
3 close all;
4
5 % =====
6 % Smart Agriculture Parameters
7 % =====
8
9 % Time (hours)
10 time = 1:10;
11
12 % Simulated soil moisture data (%)
13 soil_moisture = [30 35 28 25 20 18 22 30 40 45];
14
15 % Threshold for irrigation
16 threshold = 40;
17
18 % Pump control logic
19 % 1 = ON, 0 = OFF
20 pump_status = soil_moisture < threshold;
21
22 % Irrigation efficiency calculation (%)
23 efficiency = (soil_moisture ./ threshold) * 100;
24
25 % Limit efficiency to 100%
26 efficiency(efficiency > 100) = 100;
27
28 % =====
29 % Create Figure
30 % =====
31
32 figure;
33
34 % =====
35 % Graph 1: Soil Moisture Variation
36 % =====
37
38 subplot(3,2,1);
39
40 plot(time, soil_moisture, '-o', 'LineWidth', 2);
41
42 title('Soil Moisture Monitoring');
43 xlabel('Time (Hours)');
44 ylabel('Soil Moisture (%)');
```

